

Probability

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Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ is a collection of subsets of Ω . We call $\mathcal{F}$ a σ -algebra if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- if $(A_n)_{n \in \mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then

$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ -algebra on Ω . A function

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$$

is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{F}$,

$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A .

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ -algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω .

Remark. When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω . However, this framework will become useful when considering uncountable Ω .

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4 • *Rolling a fair die.* Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and

$$\mathbb{P}(A) = \frac{|A|}{6}$$

since all outcomes are equally likely.

- $\Omega = \{\omega_1, \dots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag.* Suppose we have n balls labelled $\{1, \dots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \dots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.

- *A deck of 52 cards.* Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.

- *Largest digit.* Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \dots, 9\}^n$, $|\Omega| = 10^n$.

Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then

$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) &= \mathbb{P}(A_k \setminus A_{k-1}) = \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$

- *Birthday problem.* There are n people. What is the probability that at least 2 of them share the same birthday?

Assume each birthday is equally likely to be one of $\{1, \dots, 365\}$. Then $\Omega = \{1, \dots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^c = \{\text{all birthdays different}\}$.

$$\begin{aligned} \mathbb{P}(A^c) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$

Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned}
 M &= \binom{n}{n_1} \cdot \binom{n-n_1}{n_2} \cdots \binom{n-(n_1+\cdots+n_{k-1})}{n_k} \\
 &= \frac{n!}{n_1!n_2!\cdots n_k!} \\
 &= \binom{n}{n_1, n_2, \dots, n_k}
 \end{aligned}$$

This is the *multinomial coefficient*.

2. Suppose $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

How many strictly increasing $f : \{1, \dots, k\} \rightarrow \{1, \dots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

What about increasing functions? Let's define a bijection

$$\{f : \{1, \dots, k\} \rightarrow \{1, \dots, n\} \text{ inc.}\} \rightarrow \{f : \{1, \dots, k\} \rightarrow \{1, n+k-1\} \text{ strictly inc.}\}$$

which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation). If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \rightarrow \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lceil x \rceil$$

Integrating from 1 to n gives

$$\begin{aligned}
 \sum_{k=1}^{n-1} \log k &\leq \int_1^n \log x \, dx \leq \sum_{k=1}^n \log k \\
 \implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\
 \implies n \log n - n + 1 &\leq l_n \leq (n+1) \log(n+1) - (n+1) + 1
 \end{aligned}$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ■

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x) \, dx = \frac{f(a) + f(b)}{2} (b-a) - \frac{1}{2} \int_a^b (x-a)(b-x) f''(x) \, dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k + 1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x \, dx = \frac{\log k + \log(k+1)}{2} + \frac{1}{2} \int_k^{k+1} \frac{(x-k)(k+1-x)}{x^2} \, dx$$

Summing from $k = 1$ to $n - 1$, we have

$$\int_1^n \log x \, dx = \frac{\log(n-1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} \, dx$. Hence

$$n \log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\begin{aligned} \log n! &= n \log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp \left(1 - \sum_{k=1}^{n-1} a_k \right) \end{aligned}$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2} \int_0^1 \frac{x(1-x)}{(x+k)^2} \, dx \\ &\leq \frac{1}{2} \int_0^1 \frac{1}{2k^2} \int_0^1 x(1-x) \, dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$\begin{aligned} A &= \exp \left(1 - \sum_{k=1}^{\infty} a_k \right) \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp \left(\sum_{k=n}^{\infty} a_k \right)}_{\rightarrow 1 \text{ as } n \rightarrow \infty} \\ \implies \frac{n!}{n^n e^{-n} \sqrt{n}} &\rightarrow A \text{ as } n \rightarrow \infty \end{aligned}$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$\begin{aligned} 2^{-2n} \binom{2n}{n} &= 2^{-2n} \frac{(2n)!}{(n!)^2} \\ &\sim \frac{2^{-2n} (2n)^{2n} e^{-2n} \sqrt{2n} A}{n^{2n} e^{-2n} n A^2} \\ &= \frac{\sqrt{2}}{A \sqrt{n}} \end{aligned}$$

Now, we shall use a different method to show

$$2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \rightarrow \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \quad (*)$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \dots = \frac{(2n-1)(2n-3)\dots 3 \cdot 1}{(2n)(2n-2)\dots 2} \cdot \frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\dots 3 \cdot 2 \cdot 1}{2n \cdot 2n \cdot (2n-2) \cdot (2n-2) \dots 2 \cdot 2} \cdot \frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{2} \cdot 2^{-2n} \binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n \dots 4 \cdot 2}{(2n+1) \dots 3 \cdot 1} \cdot 1 \\ &= \frac{1}{2n+1} \left(2^{-2n} \binom{2n}{n} \right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \rightarrow 1$ as $n \rightarrow \infty$. By (*), we have

$$\frac{I_n}{I_{n-2}} \rightarrow 1 \text{ as } n \rightarrow \infty$$

But $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$, so I_n is decreasing in n . This gives

$$\frac{I_{2n}}{I_{2n-2}} \leq \frac{I_{2n}}{I_{2n+1}} \leq \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \rightarrow \infty$,

$$\begin{aligned} \frac{\frac{\pi}{2} \cdot 2^{-2n} \cdot \binom{2n}{n}}{\frac{1}{2n+1} \cdot \left(2^{-2n} \binom{2n}{n} \right)^{-1}} &\rightarrow 1 \\ \implies \left(2^{-2n} \binom{2n}{n} \right)^2 &\sim \frac{1}{\pi n} \end{aligned}$$

■